models, pensions paid to one generation can be financed by the generations following them.[21] OLG models allow economists to measure *intergenerational inequality*: one of Raimondo's desires in reforming Rhode Island's pension plan was to reduce the subsidy paid by young workers to old workers and retirees. Modigliani won the Nobel Prize in 1985.

3.4. LIFE-CYCLE ASSET ALLOCATION

The key determinant of whether your stock allocation should shrink as you age is the correlation of human capital with equities. At a point in time, human capital is an asset, and if it is stock-like, we wish to hold a smaller equity allocation (and vice versa). The life-cycle approach takes into account how human capital changes through time.

Bond-Like Human Capital

A young person, at the start of his career, starts with relatively little financial wealth. Most of what he has is pluck, grit, and talent: human capital. If his human capital is bond-like, then his financial portfolio should offset that human capital bond position and be heavily weighted toward equities. We'll first take the extreme case and assume human capital is equivalent to risk-free bonds. Suppose that we can represent a young person's total wealth, $H+W$, comprising human capital and financial wealth by the following balance sheet:

Young Person	
Human Capital (Bonds)	$ 800,000
Financial Wealth: Stocks	$ 200,000
Financial Wealth: Bonds	$ -
Total	$1,000,000
Equity Fraction of Total Wealth	20%
Equity Fraction of Financial Wealth	100%

Suppose the young person would like a target 20% equity holding over his whole portfolio, including human capital and financial wealth. Since his human capital is equivalent to bonds, this means that he holds only stocks—and no bonds—in his financial portfolio. We can represent this balance sheet by the pie

[21] See Kotlikoff (1988) for a survey. A deep insight generated by OLG models is that population and economic growth should drive savings. If there are more young than old people, the saving of the young outweighs the dissaving of the old, and there is positive net savings. If incomes are growing, the saving of the young will outweigh the dis-saving of the old.